

Phase 4

Project Planning Phase

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Components Definition

The FitTrack AI follows a modular backend architecture in which every component performs a dedicated responsibility. The separation of concerns ensures maintainability, scalability, and efficient execution of requests. Each request passes through multiple layers before reaching the database or AI service, ensuring secure and reliable processing.

1. Client Layer (Frontend / API Consumer)

Acting as the application entry point (React web app, mobile app, or Postman), the Client Layer sends HTTP requests to the backend and renders the returned JSON responses.

- **Responsibilities:** Sends API requests, stores the JWT token post-login, invokes AI endpoints, and displays workout data alongside success or error messages.
- **APIs Accessed:** Authentication, Workout Management, AI Recommendations, and AI Fitness Insights.

2. Express Server Gateway

Acting as the central gateway, the Express server listens for incoming HTTP requests, configures middleware, and directs traffic to the appropriate controllers.

- **Responsibilities:** Starts the HTTP server, parses JSON request bodies, enables CORS, connects middleware, registers routes, and handles centralized errors.
- **Request Workflow:** Receives HTTP requests, validates request formats, forwards requests to routes, and returns structured JSON responses.

3. Authentication Middleware

The Authentication Middleware protects secured API endpoints by validating the JSON Web Token (JWT) included in the request's Authorization header.

- **Responsibilities:** Reads the Authorization header, extracts and verifies the JWT signature, decodes and attaches user data to the request object, and rejects unauthorized requests.
- **Benefits:** Prevents unauthorized access, secures workout data for authenticated users only, and eliminates repetitive authentication logic inside controllers.

4. Route Layer

The Route Layer defines the application's API endpoints and maps incoming HTTP requests to their corresponding controller methods.

- **Authentication Routes:** Register User, Login User, and User Profile.
- **Workout Routes:** Add Workout, Get All Workouts, Search Workouts, Get Workout by ID, Update Workout, and Delete Workout.
- **AI Routes:** Workout Recommendation and Fitness Insights.

5. Controller Layer

Controllers manage core application logic by receiving routed requests, validating input data, invoking services or database operations, and returning standardized JSON responses.

- **Authentication Controller:** Handles user registration, login, and profile retrieval.
- **Workout Controller:** Manages creating, retrieving, updating, deleting, and searching workout records.
- **AI Controller:** Coordinates sending prompts to Google Gemini, receiving AI-generated recommendations, and returning responses to the client.

6. Service Layer

The Service Layer encapsulates reusable business logic shared across multiple controllers.

- **Gemini AI Service:** Prepares and sends prompts to Google Gemini, processes AI-generated responses, and returns formatted recommendations.
- **JWT Service:** Generates JWT tokens, configures token expiration, and verifies user identity.
- **Password Service:** Hashes passwords prior to storage and compares credentials during login.

7. Database Layer

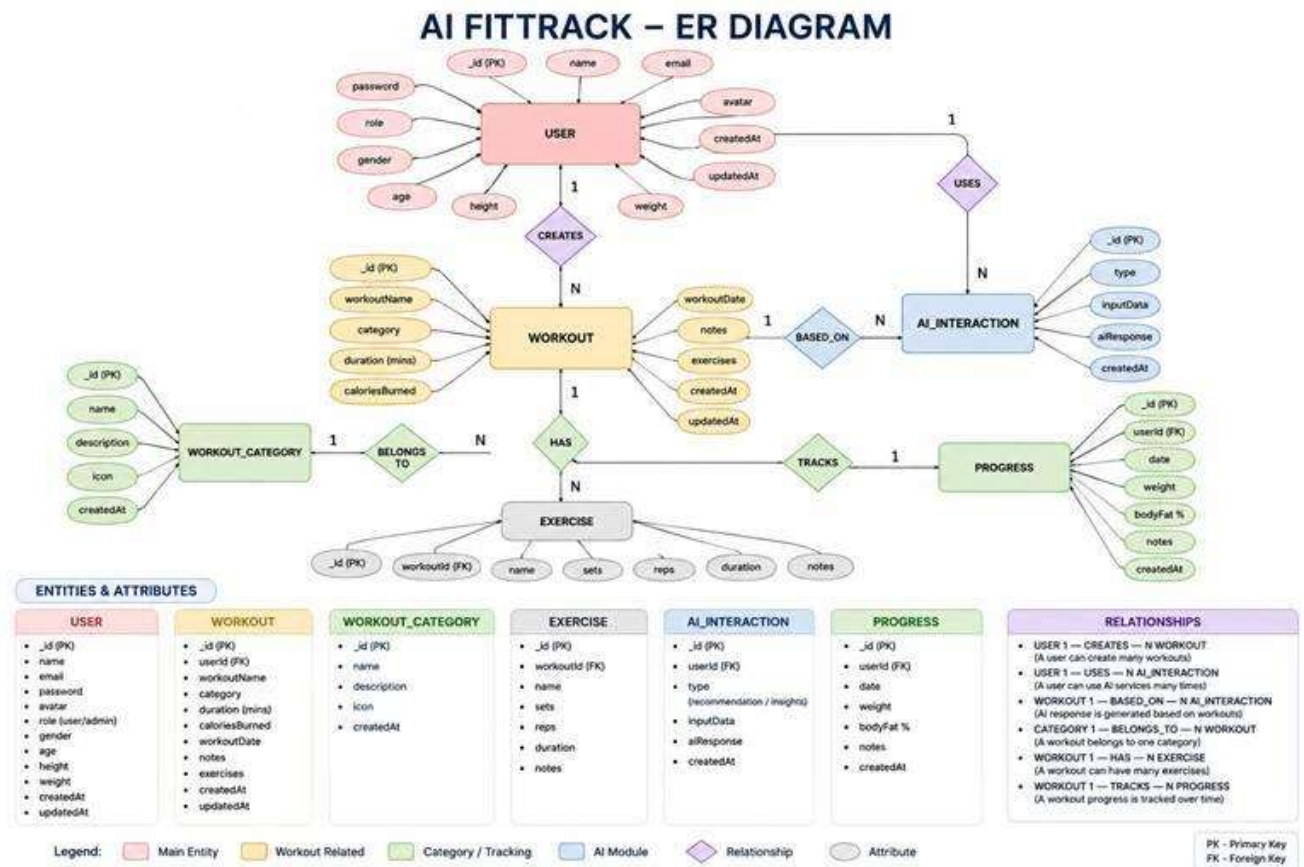
MongoDB provides persistent data storage, while Mongoose serves as the Object Data Modeling (ODM) library to streamline communication between Node.js and the database.

- **Collections:** Users Collection and Workouts Collection.
- **Advantages:** Offers a flexible schema, high performance, easy data retrieval, and a scalable architecture.

Entity Relationship (ER) Diagram

Description

The AI FitTrack backend consists of two primary collections that are logically connected through a one-to-many relationship.



1. User Entity

The User entity stores and manages user authentication credentials and profile information in the database.

- **Attributes:** Includes a primary key (_id as ObjectId), a mandatory user name (name as String), a unique email address (email as String), an encrypted password (password as String), and automated record creation and update timestamps (createdAt and updatedAt as Date).
- **Constraints & Security:** Enforces a mandatory name and unique email address, hashes passwords using bcrypt before storage, and generates a JWT token upon successful login.

2. Workout Entity

The Workout entity stores workout-related information for each authenticated user.

Attributes

Field	Type	Description
_id	ObjectId	Primary Key
user	ObjectId	References User Collection
workoutName	String	Workout Name
category	String	Workout Category
duration	Number	Duration (Minutes)
caloriesBurned	Number	Calories Burned
workoutDate	Date	Workout Date
createdAt	Date	Created Timestamp
updatedAt	Date	Updated Timestamp

Relationship

User → Workout

Relationship Type: One-to-Many

A single user can create multiple workout records. Each workout belongs to only one authenticated user.

Example: John Doe

- Morning Running
- Evening Jogging
- Upper Body Strength
- Cycling Session

This relationship is maintained using the **user** field inside the Workout schema, which stores the ObjectId of the corresponding user.

Key Features

The FitTrack AI provides a complete backend solution for workout tracking and AI-assisted fitness guidance.

1. Secure Authentication:

Implements JWT-based security encompassing user registration, secure login, password encryption, JWT generation, and protected APIs.

2. Workout Management:

Supports full CRUD operations (add, view, update, and delete workouts), with every workout record strictly linked to its authenticated user.

3. Workout Search:

Improves workout history accessibility by enabling users to search and filter records by workout name, category, or date.

4. Google Gemini Integration:

Communicates with Google Gemini AI to generate intelligent workout recommendations, taking user inputs (age, fitness goal, and experience level) to output a weekly workout plan, suggested exercises, training tips, and safety recommendations.

5. AI Fitness Insights:

Analyzes user workout statistics (total workouts, average duration, and calories burned) to produce a performance analysis, improvement suggestions, motivational advice, and a fitness progress summary.

6. Centralized Error Handling:

Returns meaningful, standardized JSON error responses for invalid credentials, missing authentication tokens, validation errors, database errors, and internal server errors.

7. RESTful API Design:

Follows REST principles supporting standard HTTP methods (**GET**, **POST**, **PUT**, and **DELETE**) and returns all responses in JSON format.

8. Modular Architecture:

Enhances maintainability and scalability by separating system responsibilities into Models, Controllers, Routes, Middleware, Services, and Configuration.

Roles and Responsibilities

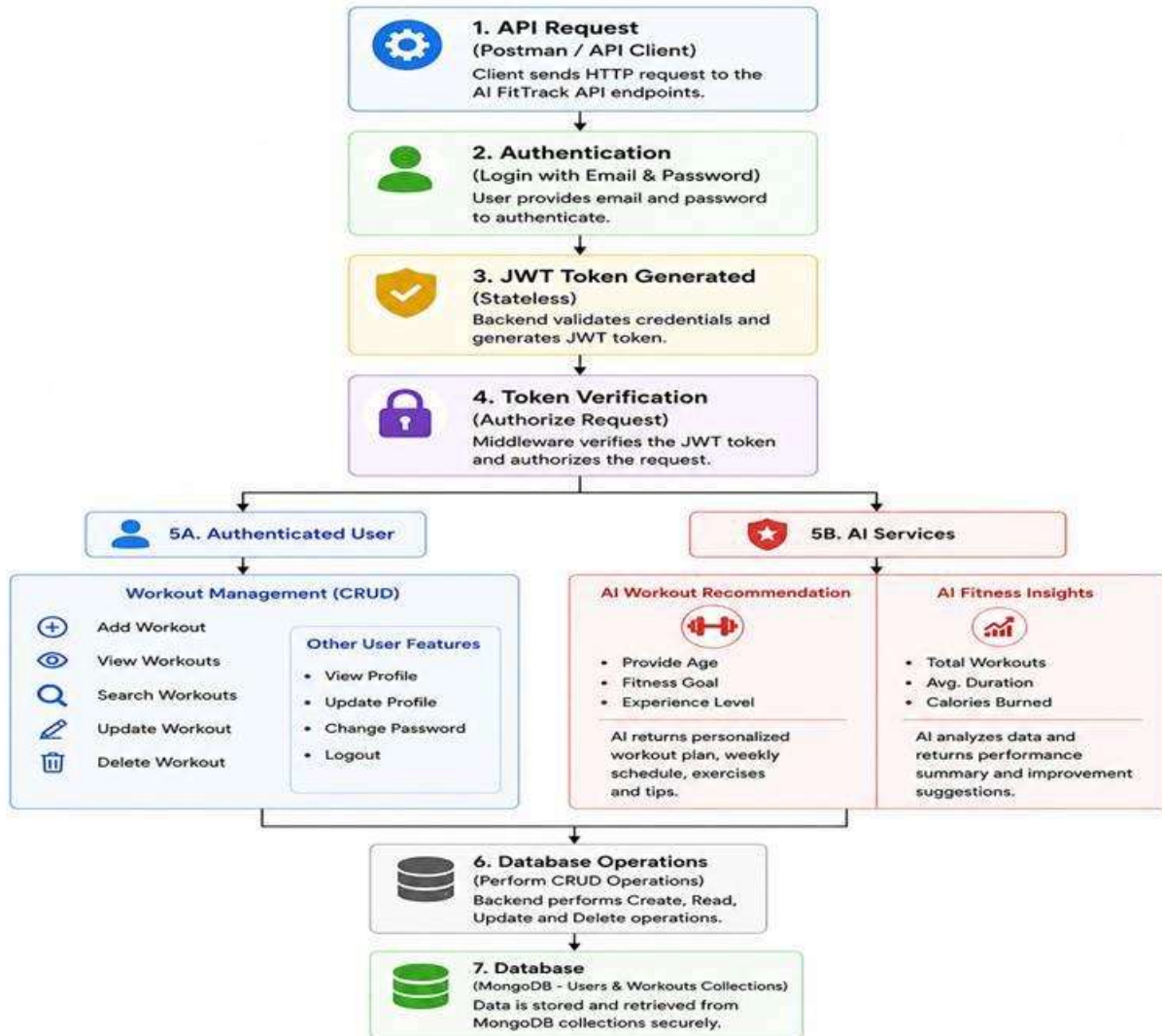
The AI FitTrack backend supports one primary role: the **Authenticated User**.

- **Responsibilities & Permissions:**

- Authenticated users hold full permissions (**Yes**) across all application operations. This authorizes them to register an account, log in securely, view profile information, perform complete workout management (add, view history, search, update, and delete workouts), and request both AI workout recommendations and AI fitness insights.

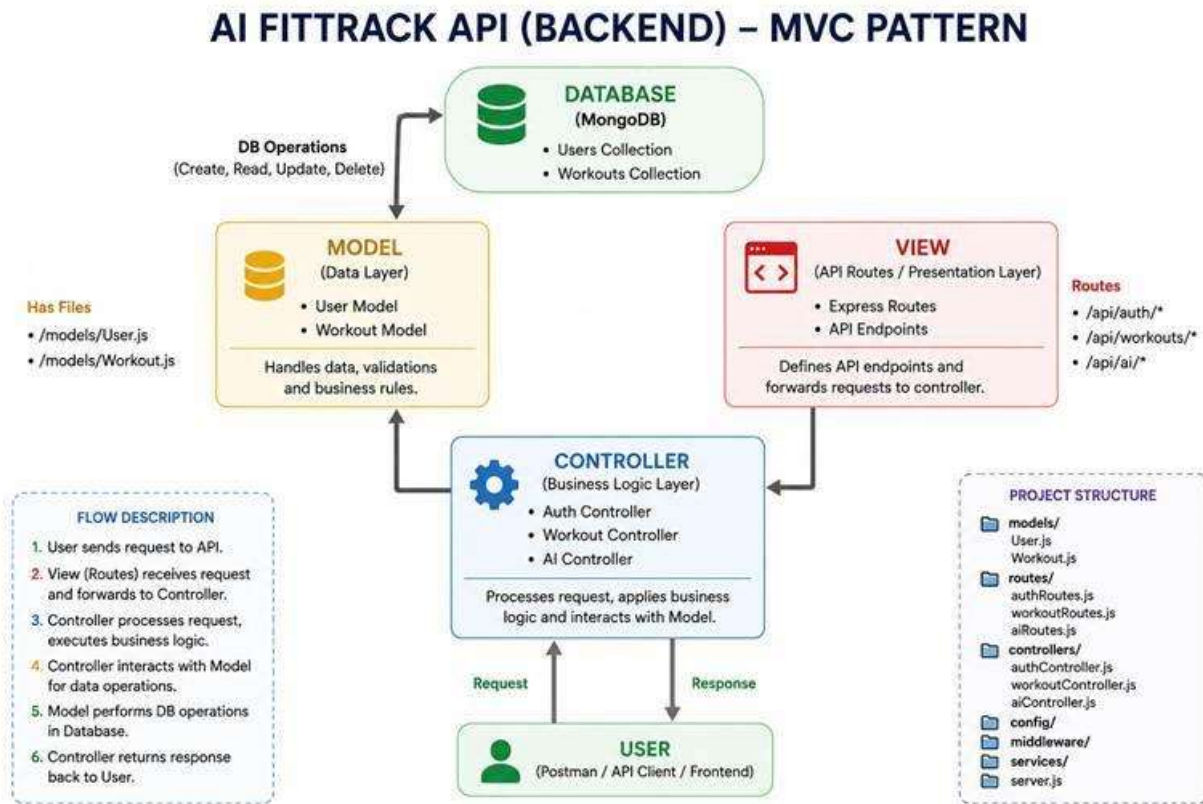
User Flow

USER FLOW DIAGRAM – AI FITTRACK API (Backend Only)



MVC Architecture Pattern

The FitTrack AI is designed using the **Model-View-Controller (MVC)** architecture pattern. This architecture separates application logic into independent components, improving code organization, maintainability, and scalability.



Model Layer (Data Layer)

The Model Layer defines the structure of the application's data using Mongoose schemas. It communicates directly with MongoDB and performs database operations.

Models in the project include:

- User Model
- Workout Model

Responsibilities:

- Schema Definition
- Data Validation
- CRUD Operations
- Collection Mapping

View Layer (API Interface)

Since AI FitTrack is a backend REST API, there are no traditional UI templates. The View Layer is represented by API consumers such as React applications, mobile applications, or Postman, which interact with the backend through HTTP requests and receive JSON responses.

Responsibilities:

- Send Requests
- Receive JSON Responses
- Display Data
- Handle Client-side Interaction

Controller Layer (Business Logic)

The Controller Layer acts as the intermediary between the View and Model. It processes incoming requests, validates input, invokes services and models, and sends structured JSON responses back to the client.

Responsibilities:

- Request Validation
- Business Logic Execution
- Database Interaction
- AI Service Invocation
- Response Generation

The separation of these three layers ensures loose coupling, simplifies testing, enhances code readability, and makes the application easier to extend with future features.